

JSS MAHAVIDYAPEETHA
JSS SCIENCE AND TECHNOLOGY UNIVERSITY, MYSURU

II Semester BE Degree Examination
Department of Mathematics
ENGINEERING MATHEMATICS-II

Duration: 3 Hours

Max. Marks: 100

Instructions:

- Part A is compulsory. Part B has internal choice.

PART – A				
Q.NO	CO	CD	PI	MARKS
1.	CO1	Comprehension	1.3.1	10
	Solve: $\frac{d^3y}{dx^3} - 6\frac{d^2y}{dx^2} + 11\frac{dy}{dx} - 6y = e^{2x} + e^{-x}$			
2.	CO2	Knowledge & Comprehension	1.2.1	10
	Find the Laplace transform of the following: (i) $\sin 3t \sin 2t$ (ii) $t \sin 4t$			
3.	CO3	Knowledge	1.1.1	10
	Find the inverse Laplace Transform of (i) $\frac{4s^5+6s^2+10s+1}{s^6}$ (ii) $\frac{3s-10}{4s^2-9} + \frac{9s-50}{16-s^2}$			
4.	CO4	Apply	1.1.1	10
	Find an approximate value of the root of the equation $x^3 - 2x - 5 = 0$ using the method of false position.			
5.	CO5	Apply	1.1.1	10
	Evaluate $\int_{-c-b-a}^c \int_{-b-a}^b \int_{-a}^a (x^2 + y^2 + z^2) dz dy dx$			

PART – B

Q.NO	CO	CD	PI	MARKS
6.	CO1	Comprehension	1.3.1	10
	Solve: $(D^2 - 4D + 4)y = e^{2x} + \cos 2x$			
OR				
7.	CO1	Comprehension	1.3.1	10
	Solve: $(D^2 + D)y = x^2 + 2x + 4$			

8.	CO2	Knowledge &Comprehension	1.2.1	10
	(i) Find the Laplace transform of $e^{at} \sin bt$ (ii) Evaluate $\int_0^{\infty} \frac{e^{-t} - e^{-3t}}{t} dt$ by using Laplace transforms.			
OR				
9.	CO2	Knowledge &Comprehension	1.1.1	10
	Derive the formula for the Laplace transform of the periodic function $f(t)$ of period T .			

10.	CO3	Knowledge &Comprehension	1.1.1	10
	Find the inverse Laplace Transform of (i) $\frac{1}{(s-3)^5}$ (ii) $\frac{1}{s(s^2-2)}$			
OR				
11.	CO3	Knowledge &Comprehension	1.1.1	10
	Find the inverse Laplace Transform of (i) $e^{-\pi s} \left(\frac{s}{s^2+9} \right)$ (ii) $e^{-2\pi s} \left(\frac{s+2}{s^2+4} \right)$			

12.	CO4	Apply	1.1.1	10
	Compute the real root of the equation $\cos x = 3x - 1$ using the Regula falsi method, correct to three decimal places (Carry out three iterations).			
OR				

13.	CO4	Apply	1.1.1	10
Use Milne's predictor-corrector method to find the value of y at $x = 0.8$, given $\frac{dy}{dx} = x - y^2$, $y(0) = 0$, $y(0.2) = 0.02$, $y(0.4) = 0.0795$, $y(0.6) = 0.1762$. Apply the corrector formula twice.				

14.	CO5	Apply	1.1.1	10
Evaluate $\iint_R xy \, dx \, dy$ where R is the region bounded by $x=0$ and $y=0$ and the line $x+y=1$				

OR

15.	CO5	Apply	1.1.1	10
Find the area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ by double integration.				

Course Outcome (CO):

CO-1	Identify second and higher order differential equations and apply to engineering problems such as oscillation of simple pendulum spring and LCR circuits.
CO-2	Apply the concept of Laplace transform and obtain Laplace transform of periodic functions and unit step functions to solve the problems in transforming the continuous signals.
CO-3	Solve problems on Inverse Laplace transforms and apply to solve linear differential equations arising from engineering problems.
CO-4	Comprehend and apply numerical techniques to solve engineering problems.
CO-5	Comprehend and apply the concepts of integral calculus.

PI

1.1.1	Apply mathematical techniques such as calculus, linear algebra and statistics to solve problems.
1.2.1	Apply laws of natural science to engineering problems
1.3.1	Apply fundamental engineering concepts to solve engineering problems

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